Vivek Kumar

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Quantitative Economics and Data Science student specializing in building end-to-end machine learning pipelines, quantitative finance algorithms, and robust data engineering architectures.

Education

• **Birla Institute of Technology, Mesra** *2022 – 2027*
Integrated M.Sc in Quantitative Economics and Data Science CGPA: 7.05

Data Science & Engineering Projects

• **Political Alpha Tracker — Quantamental Trading Engine** *Python, SQL, Streamlit, Gemini LLM, NetworkX, AWS*

- Engineered an automated quantitative trading pipeline that scrapes 5+ live data sources daily, processing over 126 companies across NSE/BSE using BeautifulSoup and yfinance.
- Constructed a complex political-corporate knowledge graph mapping Electoral Bond donations to predict government tender allocations, analyzing 130+ nodes and edges.
- Designed a multi-factor 'Conviction Score' with Kelly Criterion sizing, achieving a 56.7% win rate and +9.2% excess alpha vs Nifty 50 in backtesting.
- Developed an interactive Streamlit visualization dashboard with Gemini AI integration for natural language database querying and real-time portfolio tracking.
- Deployed the full ETL and scoring architecture via Docker and AWS EC2 with GitHub Actions CI/CD.

• **Long-Form Memory RAG** | ***Neurohack, IIT Guwahati (26th Rank)*** *FastAPI, SQLite, NLP, Embeddings*

- Secured 26th rank nationally (Certificate) by architecting a highly durable context-memory system for LLMs capable of handling 1,000+ conversation turns without prompt decay.
- Designed a hybrid retrieval strategy combining semantic cosine similarity with lexical BM25 ranking via SQLite FTS5, ensuring sub-100ms context injection latency.
- Built asynchronous extraction pipelines to continuously map, update, and supersede conversational memory states in the background.

• **Heuristic Optimization via Genetic Algorithms** *Python, SciPy, Pandas, NumPy*

- Engineered a custom heuristic machine learning algorithm (Real-coded Genetic Algorithm) from scratch to solve high-dimensional, non-convex optimization problems.
- Constructed robust data pipelines using Pandas to preprocess time-series data, conducting rigorous out-of-sample backtesting to validate algorithmic superiority over baseline models.

• **Bayesian Inference & MCMC Modeling** *R, Probabilistic Programming, Statistics*

- Implemented a Metropolis-Hastings (Markov Chain Monte Carlo) algorithm in R to perform robust Bayesian inference on early-stage COVID-19 transmission rates.
- Estimated critical epidemiological parameters by fitting complex probabilistic models to real-world healthcare datasets.

Research & Publications

- **Vivek Kumar**, Dr. Mrinal Jana. "Distribution-free Uncertainty Quantification for Option Pricing Using Residual-space Conformal Prediction." *CIMA 2026*, **Accepted**. Proceedings to appear in Springer *LNNS* (Paper ID-498, Scopus-indexed).

Technical Skills

Languages: Python, R, SQL, C++
Data Science & ML: Scikit-learn, XGBoost, TensorFlow, PyTorch, Keras, Statsmodels, NLP, Feature Engineering
Data Engineering: Pandas, NumPy, SQLite, Web Scraping (BeautifulSoup), API Integration, ETL Pipelines
Cloud & DevOps: Docker, Git, GitHub Actions (CI/CD), AWS EC2, Linux, FastAPI
Data Visualization: Matplotlib, Seaborn, Plotly
Relevant Coursework: Probability, Statistical Inference, Econometrics, Time Series Analysis, Machine Learning, Multivariate Data Analysis, Optimization, Algorithms for Big Data

Leadership & Extracurricular Activities

• **Events & Public Relations Organizer** | ***Data Science Summit '23, BITOTSAV '24***

- Spearheaded public relations and corporate outreach for the Data Science Summit (DSS 23), successfully pitching and securing key event sponsorships.
- Organized logistics and event management for Pantheon 2023 and BITOTSAV 2024 college festivals.